

Ibtesam Hussain

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PROFILE

Software Engineer with strong foundations in **object-oriented programming, data structures, and Full-Stack development**. Experienced in building scalable APIs and integrating AI-driven features into applications. Looking forward about developing efficient, production-ready software systems.

EXPERIENCE

Aykeys Startup — AI Software Engineer Intern (Onsite, Karachi, Pakistan)

July 2025 — Sept 2025

- Developed and integrated AI-powered features using **LLMs (GPT-4.1, Mistral 7B) via LangChain and OpenRouter APIs** to enhance application functionality and response quality.
- Designed and implemented backend APIs using Flask for serving AI/ML inference, enabling seamless integration with web applications.
- Built and contributed to multiple **full-stack applications**, including an AI Portfolio Generator, Q&A Forum, Product Catalog, and Booking System using **React, Node.js, Firebase, and MongoDB**.
- Collaborated in an **agile development** environment using **Git/GitHub** for version control and tools like Slack and Asana for **SLDC Lifecycle** and team communication.

FAST NUCES — Final Year Project Lead (Hybrid, Karachi, Pakistan)

Sept 2025 — May 2026

- Led a **3-member research team**, delivering a **novel hybrid monocular depth estimation system for clinical wound imaging**.
- Designed a two-pipeline system** combining Gemini AI-synthesized stereo viewpoints with DA3 Giant monocular depth model, fused via Laplacian pyramid technique.
- Achieved **43.1%** of image pairs reaching **calibrated stereo range** with best **reprojection errors of 5.26 gray levels** across the dataset.

SKILLS

Programming Languages: C/C++, Java, Python, JavaScript, TypeScript, SQL, C#.

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems (SQL), Software Design Principles.

Software Development Tools: Git/GitHub, Jira, Slack, Jupyter Notebook, Google Colab

Back-end & Deployment: RESTful APIs, Flask, FastAPI, Node.js, .NET, Model Inference APIs, Cloud Deployment (AWS, GCP, Vercel, Render, Streamlit)

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Deep Learning (CNNs, RNNs, LSTM), Transfer Learning, YOLO

PROJECTS

Rent a Car — (Java, OOP-based Application)

- Developed a Desktop-based **car rental management application** using **Java**, applying core **object-oriented programming principles including encapsulation, inheritance, and polymorphism**.
- Designed modular components for managing vehicles, customers, bookings, and billing operations within the system.
- Implemented business logic for rental workflows such as vehicle availability tracking, booking validation, and cost calculation.
- Utilized appropriate data structures to efficiently manage and retrieve system data during runtime.
- Built a GUI to enable smooth user interaction and navigation across system functionalities.

Music Web App — (MERN – Stack Web Application)

- Established a full-stack music web application using MongoDB, Express.js, React.js, and Node.js.
- Managed **50+ songs** with audio, cover images, and metadata in MongoDB using a custom seed.js script.
- Designed interactive UI with **100+ artists**, real-time audio controls, and responsive Tailwind CSS layout.
- Implemented user-specific favorites and global audio state management across components.
- Made **user authentication** and **role-based access** control with secure login, registration, and session Management.

Community Forum — (WebApp, CI/CD, Parallelism)

- Build a community forum web application using **React for the frontend** and **Firebase (GCP) for the backend**, enabling users to discuss topics, ask questions, and share knowledge in a collaborative environment.
- Configured CI/CD workflows** via **GitHub Actions** to automate build checks and streamline deployments to Firebase Hosting.
- Implemented logging and basic monitoring for backend services to support real-time debugging and uptime visibility.
- Fix multiple failed CI/CD deployments and **ESLint issues**.
- Maintain **concurrency** and **parallelism** among users.

ConsciousDay Agent — (AI Powered Web Application, LLM Integration)

- Developed an **AI-powered** journaling and productivity web application using **DeepSeek R1 LLM** to generate personalized reflections and recommendations.
- Designed and implemented user **data management and context-aware memory** handling, enabling retrieval of historical entries and improving response relevance.
- Built features for tracking and visualizing user productivity trends based on historical data and AI-generated insights.
- Developed and deployed the application using **Streamlit**, ensuring persistent user data handling and smooth user interaction.

EDUCATION

FAST National University of Computer and Emerging Sciences (NUCES), Karachi

August 2022 — June 2026

Bachelor of Science in Artificial Intelligence | 2x Dean's Honors List

Relevant Courses: Machine Learning, Deep Learning, Artificial Intelligence, Computer Vision, Agentic AI, DSA, OOPs, NLP.